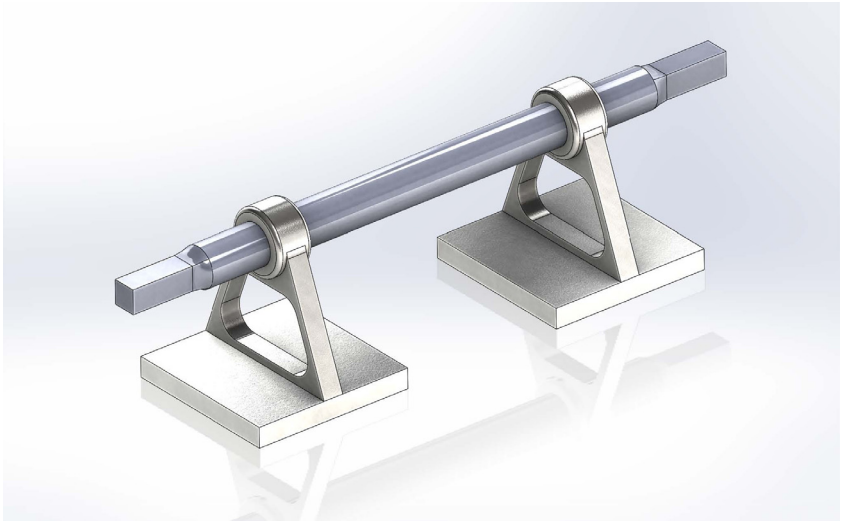
Lab 1
SolidWorks – Assembly and Contact Simulations

Objective

The objective of this lab is to become familiar how to run a simulation study of an assembly involving displacement contact.

Procedure

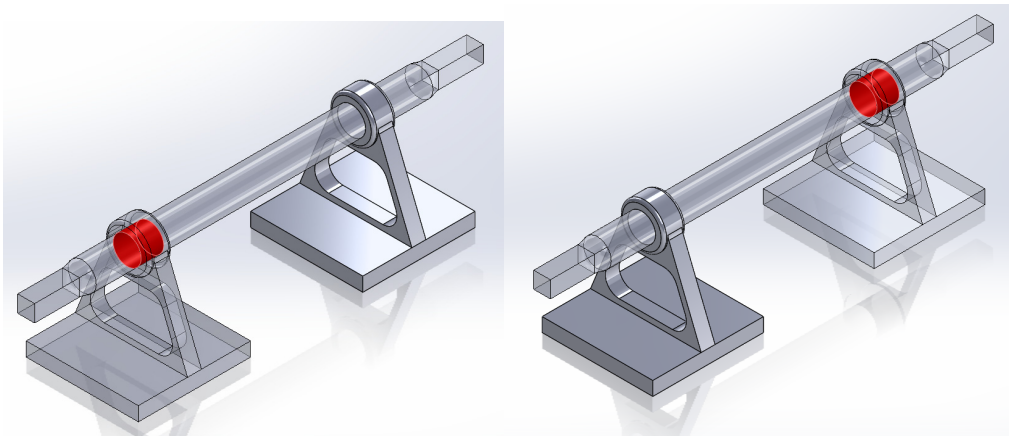
The assembly includes three parts; two bearings and a shaft that are previously designed. The shaft is ideally fit in between the two bearings. One important step before running a FEA analysis on an assembly of parts is use the SolidWorks interference detection tool to find out whether there are any actual interferences. Interference detection tool can be accessed in the assembly mode through `Evaluate>Interference Detection`. After importing the parts to the assembly, as shown below, and running the interference tool, in the options check the `Treat coincidence as interference.`



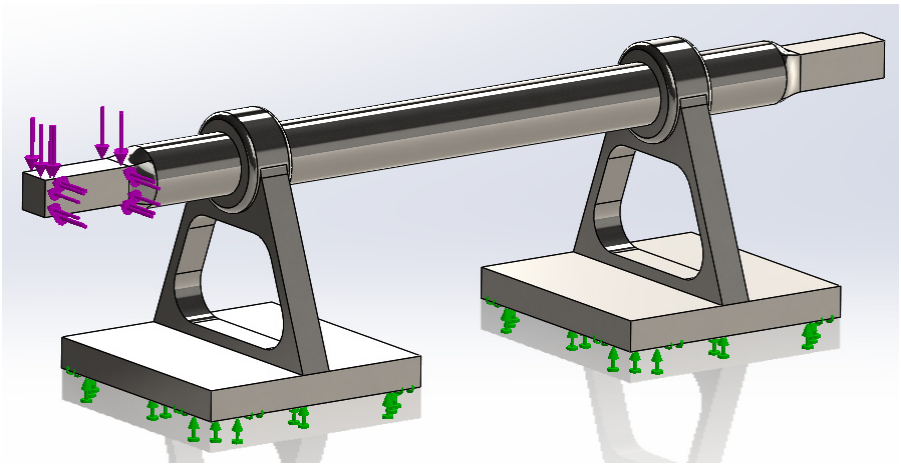
The `Treat coincidence as interference` will help to identify

- The interfaces
- Touching faces

After running the interference check, two coincident interferences should show up in the “Results”. This also indicates that we do not have any actual interference faces. This is required since if there are interferences between parts that will affect meshing and may impede the ability to solve a problem. However, these interferences are just coincidences only (as shown below).

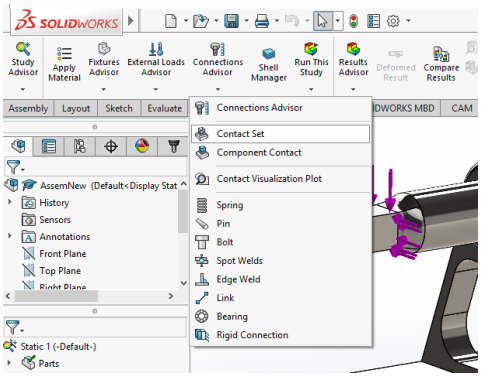


There are two sets of touching faces. After assigning individual materials (e.g., apply AISI 304 Steel to all the parts), apply fixtures (Fixed geometry constraints) on each of the bearings and loading (200lb each) downward and sideways on the endpoint of the shaft as shown.

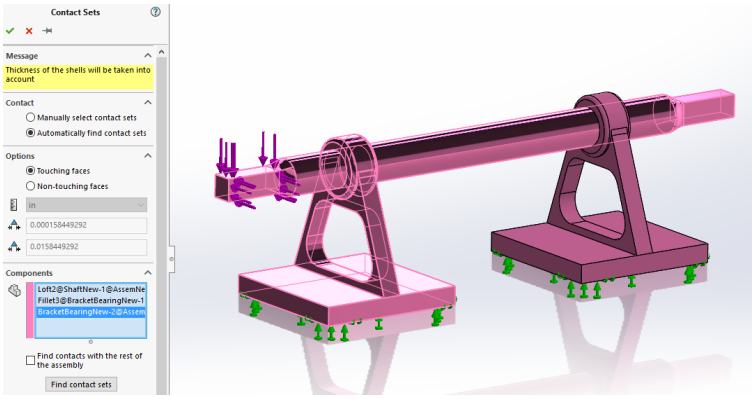


It is needed to define how the contact will be handled between those touching faces. By default, in the assembly, SolidWorks creates a global contact condition under:
`Connections>Component Contacts>Global Contact (-Bonded-).`

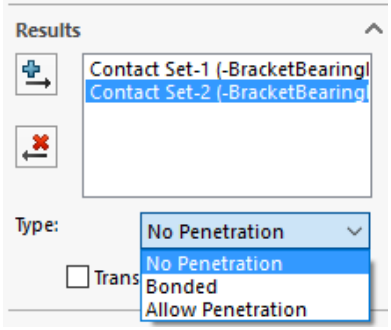
And the default is generally set to bonded. This means any part that is touching any other parts will be treated as bonded, unless if there are any local contact sets that have been defined. In our case, we would like to define local contact sets that will maintain contact for the shaft to the bearings but will allow them to separate if the load become significant enough. This can be done using the connection commands that can be found via
`Simulation>Connection Advisor>Contact Set`



In the contact set command, contact sets can be defined manually by matching the pair of faces that are expected to come in contact. Alternatively, SolidWorks can automatically find those contact sets based on picking on the desired components (i.e., two bearings and the shaft). This can be done by clicking on such parts as they appear in the components list.



By choosing `Find faces`, it determines two contact sets that are the exact sets we found when we ran the interference detection. Set them to a `No Penetration` contact. Therefore, SolidWorks will monitor the contact between these faces – depending on the loading condition, theses faces may separate from each other. Now, Mesh the model and run the simulation. Note that for analyses involving no penetration contacts much longer runtimes should be expected.



Lab write-up:

In your report, select a portion of your final "design project" where you can apply the features learnt from this manual. Compare your results in detail with a case when face penetration is allowed.